Solution:
Let $P(n) := \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$. Substituting the value of $n = 1$, in the statement we get, $P(1) = \frac{1}{1.2} = \frac{1}{2}$. Hence, $P(1)$ is true. Let us assume that the statement is true for $n = k$. Then

$$P(k) = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{k(k+1)} = \frac{k}{k+1}.$$

We need to show that $P(k+1)$ is true. Consider,

$$\begin{aligned} P(k+1) &= \underbrace{\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{k(k+1)}} + \frac{1}{(k+1)(k+2)} \\ &= P(k) + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{1}{k+1}\left(\frac{k}{1} + \frac{1}{k+2}\right) \\ &= \frac{1}{k+1}\left(\frac{k^2+2k+1}{k+2}\right) \\ &= \frac{1}{k+1}\left(\frac{(k+1)^2}{k+2}\right) = \frac{(k+1)}{(k+2)}. \end{aligned}$$

This implies, $P(k+1)$ is true. The validity of $P(k+1)$ follows from that of $P(k)$. Therefore, by the principle of mathematical induction, for any natural number n,

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

Example 4.65 Prove that for any natural number n, $a^n - b^n$ is divisible by $a - b$, where $a > b$

Solution:
Let

$$P(n) := a^n - b^n, \text{ is divisible by } a - b.$$

Substituting the value of $n = 1$, in the statement we get,

$$P(1) = a - b,$$

which is divisible by $a - b$. Hence, $P(1)$ is true. Let us assume that the statement is true for $n = k$. Then $P(k) = a^k - b^k$, is divisible by $a - b$. We can write

$$P(k) = a^k - b^k = \lambda(a - b),\ \lambda \in \mathbb{N}.$$

We need to show that $P(k+1) = a^{k+1} - b^{k+1}$, is divisible by $a - b$.

$$\begin{aligned} P(k+1) &= a^{k+1} - b^{k+1} \\ &= a^{k+1} - ab^k + ab^k - b^{k+1} \\ &= a(a^k - b^k) + b^k(a - b) \\ &= a(\lambda(a - b)) + b^k(a - b) \\ &= (a - b)\left(a\lambda + b^k\right) \\ &= (a - b)\lambda_1,\ \lambda_1 = a\lambda + b^k,\ \lambda_1 \in \mathbb{N}. \end{aligned}$$